Rank-one degree entropy does not add a budget in the classical monopole family

23 September 2026. This is a composition of classical mapping rigidity, canonical-divisor positivity, and curvature inequalities. It closes one specific proposed route to an independent budget; it is not a new Einstein nonexistence theorem. It applies to the entire classical family below, not only to the ball-quotient examples used in the two-chirality application.

1. Statement

Let

$$M = X\#k\overline{\mathbb{CP}}^2\#m(S^1\times S^3), \qquad A = c_1^2(X) > 0, \quad h = (k+4m)/A,$$

where X is a closed minimal complex surface of general type and $k, m \geqslant 0$. Give X its complex orientation. Then:

1. Every continuous map from M to a closed real-hyperbolic four-manifold has degree zero.
2. If $f : M \to Y$ maps to a closed complex-hyperbolic surface and $D = |\deg f| > 0$, then

$$\boxed{D\,c_1^2(Y) \leqslant A.} \tag{1}$$

3. For every metric whose Ricci eigenvalues lie in $[-1-\delta, -1+\delta]$, with $0 \leqslant \delta < 1$, the classical scalar monopole bound gives

$$\boxed{\mathrm{Vol}(M,g) \geqslant \frac{2\pi^2 A}{(1+\delta)^2}.} \tag{2}$$

   Every ordinary degree-to-complex-hyperbolic BCG/Bishop lower bound is no larger than 64/81 times the right side of (2).

Consequently, the ordinary rank-one target degree inequalities do not strengthen the available volume lower budget for pointwise negative Ricci input on this family. For exact negative Einstein candidates their energy and Chern–Weil consequences are already weaker than the classical monopole Ricci-energy bound. This is a comparison of these specified inequalities, not a claim that all entropy, representation, or harmonic-map methods have been exhausted.

2. Why all degree comes from the Kähler summand

For any aspherical target Y, pinch the connected-sum necks to obtain

$$q : M \longrightarrow X \vee \bigvee_1^k \overline{\mathbb{CP}}^2 \vee \bigvee_1^m (S^1\times S^3).$$

This map induces the free-product isomorphism on fundamental groups. Maps to $Y = K(\pi_1 Y, 1)$ are classified, up to free homotopy, by fundamental-group homomorphisms up to conjugation. Thus every f is homotopic to $F \circ q$ for a map on the wedge, obtained by restricting its fundamental-group homomorphism to the summands. The fundamental class is sent by q to the sum of their fundamental classes, so degrees add.

The simply connected projective-plane summands map null-homotopically. A map from $S^1 \times S^3$ to Y is homotopic to one factoring through S^1, since this realizes the same fundamental-group map.

These summands therefore have degree zero. There is a continuous map $f_X : X \to Y$ satisfying

$$\deg f_X = \deg f. \tag{3}$$

This argument does not presume that the unknown metric on M is Kähler or that an embedded copy of the punctured X is geometrically adapted to it.

Minimal surfaces of general type are projective and hence Kähler. Give X any Kähler metric and replace f_X by its harmonic representative into the closed nonpositively curved target. For a real-hyperbolic target, Sampson's rank theorem gives rank at most two, so its four-dimensional degree is zero. For a complex-hyperbolic target, a nonzero degree forces real rank four somewhere; harmonic rigidity then makes the map holomorphic or antiholomorphic. These precise alternatives are recorded in Carlson–Toledo, Corollary 3.7; harmonic representative existence is also used there. In complex dimension two conjugating the target complex structure preserves its real orientation and its canonical square.

3. The degree bound is a canonical intersection calculation

We can therefore assume $f_X : X \to Y$ is holomorphic of degree D. It is surjective and generically finite. The determinant of its differential gives the canonical divisor identity

$$K_X = f_X^* K_Y + R,$$

where the ramification divisor R is effective. The canonical class K_X is nef because X is minimal of general type; K_Y is ample, so $f_X^* K_Y$ is nef. It follows that

$$A - D c_1^2(Y) = K_X^2 - (f_X^* K_Y)^2 = R \cdot (K_X + f_X^* K_Y) \geqslant 0.$$

This proves (1), including maps with ramification and exceptional fibres. No assumption that f_X is an unramified covering is used.

4. Compare the budgets at the actual pointwise normalization

Normalize the complex-hyperbolic metric g_Y by $\mathrm{Ric}_{g_Y} = -g_Y$, and write $B = c_1^2(Y)$. Then

$$V_Y = 2\pi^2 B, \qquad h_{\mathrm{vol}}(g_Y)^4 = 64/9.$$

For example, the holomorphic-sectional-curvature -4 normalization has Ricci $-6g$ and entropy 4; scaling by six gives the displayed values. Equivalently they give exactly the 32/81 coefficient in Kotschick's complex-hyperbolic degree obstruction.

BCG's degree inequality and Bishop comparison under $\mathrm{Ric}_g \geqslant -(1+\delta)g$ give

$$h_{\mathrm{vol}}(g)^4 V \geqslant (64/9) D V_Y, \qquad h_{\mathrm{vol}}(g)^4 \leqslant 9(1+\delta)^2.$$

Hence the lower number furnished by this method is

$$\boxed{V \geqslant \frac{64}{81} \frac{2\pi^2 D B}{(1+\delta)^2}.} \tag{4}$$

On the other hand, the polarized actual monopole class on M has $(a_g^+)^2 \geqslant A$, including the classical small-b_+ chamber and handle-insertion hypotheses supplied in the inherited closure proof. The scalar inequality gives

$$32\pi^2 A \leqslant \int s^2 \leqslant 16(1+\delta)^2 V,$$

which is (2). By (1), the right side of (4) is at most 64/81 of the right side of (2). This comparison is uniform over every target and every degree map. An arbitrary collection of such lower bounds can be maximized, not added; its supremum remains dominated.

For any normalized exact negative Einstein metric put $e = \int |\mathrm{Ric}|^2/(8\pi^2 A) = V/(2\pi^2 A)$. Equation (4) reads $e \geqslant (64/81)DB/A \leqslant 64/81$. The classical all-metrics Ricci bound is stronger:

$$e \geqslant 1 + h. \tag{5}$$

Indeed, the scalar–Weyl defect $\mathscr{D}_{\mathrm{LB}} \geqslant 0$ gives $\int |\mathrm{Ric}|^2 = 8\pi^2(2p-T)+32\pi^2\mathscr{D}_{\mathrm{LB}}$, with $p \geqslant A$, $T = A(1-h)$. Thus (5) does not depend on attainment of an energy infimum.

In the range $0 < h < 1/3$, the stronger classical Einstein envelope $e \geqslant F(h) > 1 + h$ already implies all these entropy lower bounds. In terms of the two-chirality upper budget U, Kotschick's necessary degree inequality is $U > (64/81)DB/A$; any candidate with $U \geqslant F(h)$ automatically passes it. It therefore cannot close the remaining region $U - F(h) > 0$ by this particular comparison.

5. Scope and research consequence

This note provides a family-wide reason to stop repeatedly seeking the missing budget from ordinary four-dimensional rank-one target degree comparisons on the classical family. The result includes the realized ball-quotient cover examples, but is not limited to them.

It does not cover Bauer–Furuta connected sums of non-Kähler symplectic summands: the Kähler source and nef canonical argument above would have to be replaced. It also does not bound the manifold's actual minimal entropy from above, rule out additional quantitative stability inputs, or concern targets of higher rank, representations without ordinary degree maps, noncompact targets, or orbifold degree transfers. It supplies no new pointwise approximation and no new no-Einstein type. The unrestricted integral dilution examples in Round 18 and the real-hyperbolic surgery comparison in Round 19 remain different results.

Sources and finite checks

- Carlson–Toledo, *Harmonic mappings of Kähler manifolds to locally symmetric spaces*, IHÉS 69 (1989), Corollaries 3.3–3.7: https://www.numdam.org/item/10.1007/BF02698844.pdf.
- Besson–Courtois–Gallot, *Entropies et rigidités des espaces localement symétriques de courbure strictement négative*, GAFA 5 (1995), main theorem: https://www.aimath.org/WWN/entropyrigidity/papers/BessonCourtoisGallot.pdf.
- Kotschick, *Entropies, volumes, and Einstein metrics*, Corollary 3, equation (15): https://arxiv.org/html/math/0410215.
- LeBrun, *Ricci Curvature, Minimal Volumes, and Seiberg–Witten Theory*, Proposition 3.2 and the Ricci-energy inequality: https://arxiv.org/html/math/0003068.

`check_line1_rank_one_entropy_scope.py` checks the scale factors, the volume/energy domination, and the translation to Kotschick's coefficient. It does not check the harmonic-map or Seiberg–Witten existence theorems.